

Expt. No. : 8**Date :****AGGREGATION USING NUMPY****AIM**

To write a python program to implement aggregation functions using numpy packages.

SOFTWARE REQUIRED

Python 3.14

THEORY

NumPy is a powerful library in Python for numerical and mathematical operations, and it provides various aggregation functions to perform operations on arrays. Aggregation functions in NumPy allow you to perform computations across the entire array or along a specified axis. Here are some commonly used NumPy aggregation functions:

VARIOUS AGGREGATE FUNCTIONS

Function Name	Description	Example Usage
np.sum()	Sum of array elements	np.sum(array)
np.mean()	Mean of array elements	np.mean(array)
np.median()	Median of array elements	np.median(array)
np.std()	Standard deviation of array elements	np.std(array)
np.var()	Variance of array elements	np.var(array)
np.min()	Minimum value in array	np.min(array)
np.max()	Maximum value in array	np.max(array)
np.argmin()	Index of minimum value in array	np.argmin(array)
np.argmax()	Index of maximum value in array	np.argmax(array)
np.percentile()	Percentile (e.g., 50th percentile as median)	np.percentile(array, 50)
np.any()	Evaluates if any elements are true	np.any(array > 5)
np.all()	Evaluates if all elements are true	np.all(array > 0)

PROBLEM

1. Build a Python program using NumPy to determine the product of all elements in a one-dimensional array and compare it with the sum of the elements.
2. An electrical maintenance department continuously monitors the temperatures of transformers, motors, and switchgear. Build a Python program using NumPy to analyze the temperature readings by determining the average, highest, lowest, variance, and standard deviation.
3. You are given a space separated list of numbers. Your task is to print a reversed NumPy array with the element type float.

ALGORITHM

STEP 1: Start the program

STEP 2: import numpy as np

STEP 3: Create get_user_array() and perform_aggregation(array) function

STEP 4: Function get_user_array()

- Get the values of rows and columns of array from the user and the array elements, else if ValueError occurs, ask the user to enter a numeric value
- return the array

STEP 5: Function perform_aggregation :

- Check if the array is not empty, If true: Calculate the sum, mean, maximum, minimum of the array using the respective numpy aggregate functions and the print the values

STEP 6: In Main() function:

- Get the value of array using get_user_array() function
- Call the perform_aggregation function and pass the array as argument

STEP 7: Call the main function

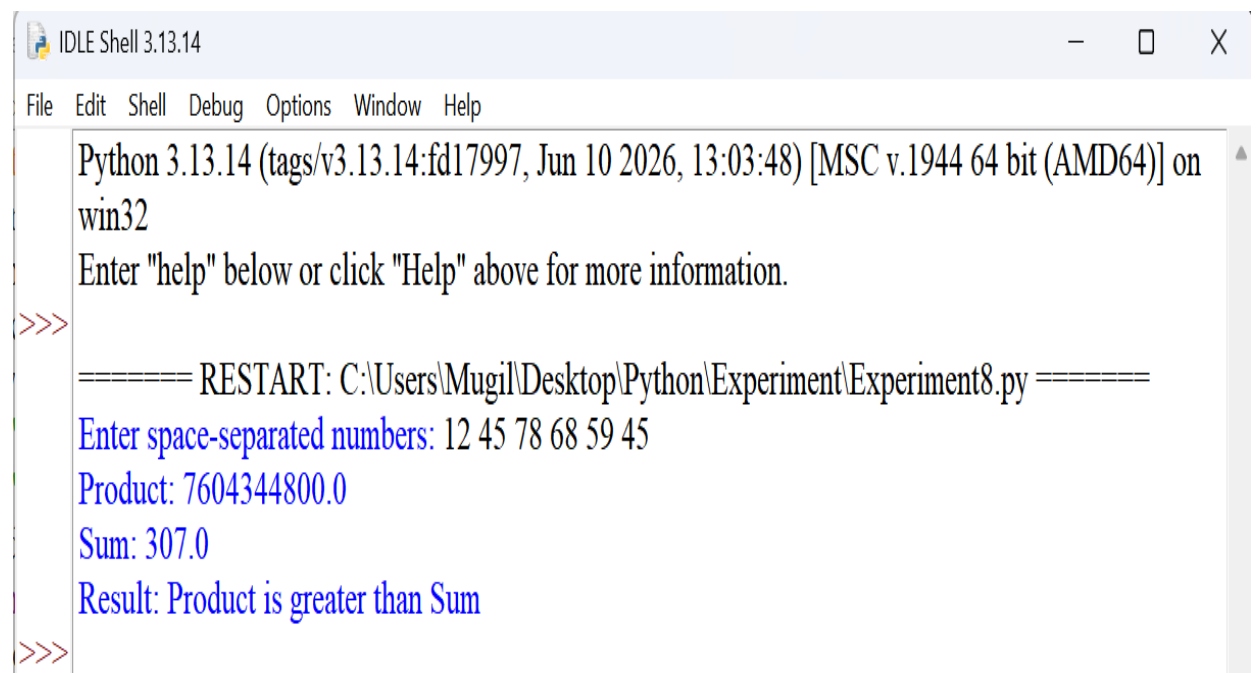
STEP 8: Terminate the program.

PROGRAM CODING

```

import numpy as np
elements = list(map(float, input("Enter space-separated numbers: ").split()))
arr = np.array(elements)
product_val = np.prod(arr)
sum_val = np.sum(arr)
print("Product:", product_val)
print("Sum:", sum_val)
if product_val > sum_val:
    print("Result: Product is greater than Sum")
elif product_val < sum_val:
    print("Result: Sum is greater than Product")
else:
    print("Result: Product and Sum are equal")

```

OUTCOMES OF PROGRAM CODING


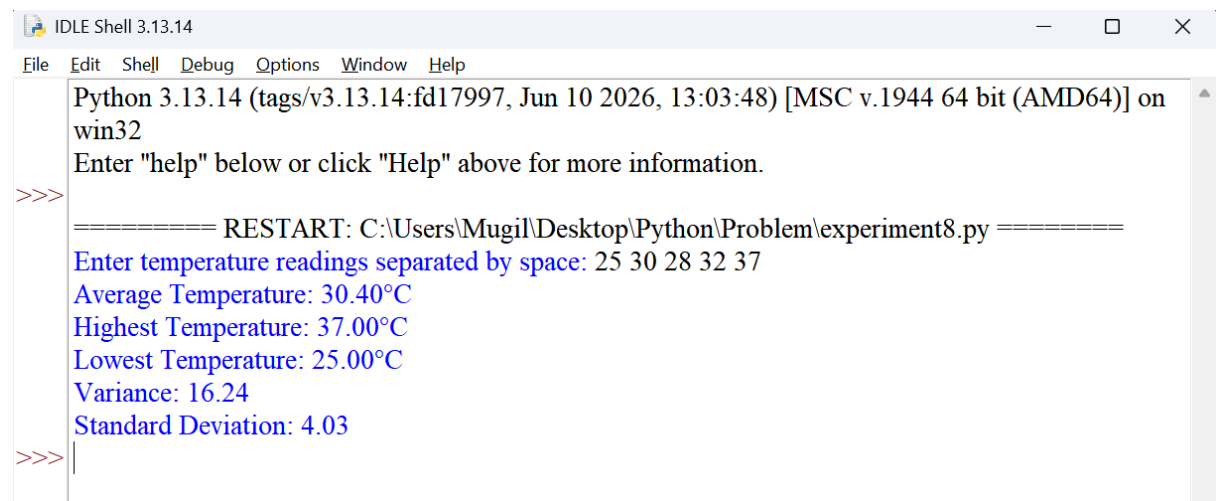
```

IDLE Shell 3.13.14
File Edit Shell Debug Options Window Help
Python 3.13.14 (tags/v3.13.14:fd17997, Jun 10 2026, 13:03:48) [MSC v.1944 64 bit (AMD64)] on win32
Enter "help" below or click "Help" above for more information.
>>>
===== RESTART: C:\Users\Mugil\Desktop\Python\Experiment\Experiment8.py =====
Enter space-separated numbers: 12 45 78 68 59 45
Product: 7604344800.0
Sum: 307.0
Result: Product is greater than Sum
>>>

```

PROBLEM CODING

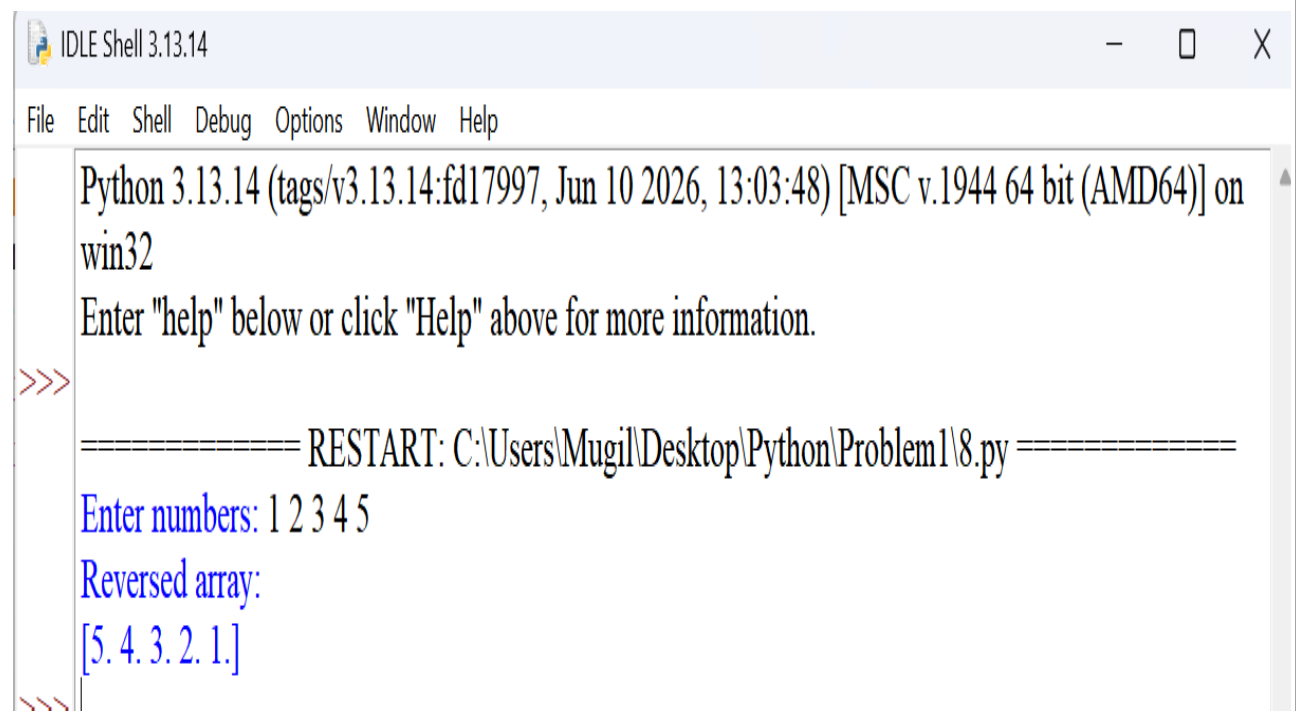
```
import numpy as np
temps = np.array(list(map(float, input("Enter temperature readings
separated by space: ").split()))))
print(f"Average Temperature: {np.mean(temps):.2f}°C")
print(f"Highest Temperature: {np.max(temps):.2f}°C")
print(f"Lowest Temperature: {np.min(temps):.2f}°C")
print(f"Variance: {np.var(temps):.2f}")
print(f"Standard Deviation: {np.std(temps):.2f}")
```

OUTCOMES OF PROGRAM CODING

```
IDLE Shell 3.13.14
File Edit Shell Debug Options Window Help
Python 3.13.14 (tags/v3.13.14:fd17997, Jun 10 2026, 13:03:48) [MSC v.1944 64 bit (AMD64)] on
win32
Enter "help" below or click "Help" above for more information.
>>>
===== RESTART: C:\Users\Mugil\Desktop\Python\Problem\experiment8.py =====
Enter temperature readings separated by space: 25 30 28 32 37
Average Temperature: 30.40°C
Highest Temperature: 37.00°C
Lowest Temperature: 25.00°C
Variance: 16.24
Standard Deviation: 4.03
>>>
```

PROBLEM CODING

```
import numpy as np
numbers = input("Enter numbers: ").split()
arr = np.array(numbers, dtype=float)
print("Reversed array:")
print(arr[::-1])
```

OUTCOMES OF PROGRAM CODING

```
IDLE Shell 3.13.14
File Edit Shell Debug Options Window Help
Python 3.13.14 (tags/v3.13.14:fd17997, Jun 10 2026, 13:03:48) [MSC v.1944 64 bit (AMD64)] on win32
Enter "help" below or click "Help" above for more information.
>>>
=====RESTART: C:\Users\Mugil\Desktop\Python\Problem1\8.py=====
Enter numbers: 1 2 3 4 5
Reversed array:
[5. 4. 3. 2. 1.]
>>>
```

VIVA QUESTION**1. What is aggregation in the context of NumPy?**

Aggregation refers to the process of summarizing data by performing operations such as calculating mean, sum, or count across an array, reducing multiple values to a single value

2. How do you calculate the mean of an array using NumPy?

Use the `mean()` function to calculate the average of the elements in a NumPy array.

3. What function would you use to compute the standard deviation of an array in NumPy?

The `std()` function is used to compute the standard deviation of an array in NumPy.

4. Explain the difference between `mean()` and `average()` in NumPy.

`mean()` and `average()`:

The `mean()` function calculates the arithmetic mean, while `average()` can compute a weighted average if weights are provided; otherwise, it behaves similarly to `mean()`.

5. How can you calculate the sum of all elements in a NumPy array?

Use the `sum()` function to calculate the sum of all elements in a NumPy array.

6. What is the purpose of the `axis` parameter in NumPy aggregation functions?

The `axis` parameter specifies the dimension along which to perform the aggregation. For example, it can aggregate rows or columns in a multi-dimensional array.

7. How do you compute the median of an array using NumPy?

Use the `median()` function to compute the median of an array in NumPy.

8. Can you calculate the variance of a dataset using NumPy? If so, how?

Yes, you can calculate the variance of a dataset using the `var()` function in NumPy.

9. How does NumPy handle NaN values during aggregation?

NumPy functions typically ignore NaN values in aggregation operations, but you can use functions with specific parameters to handle NaNs as needed.

10. Explain how you would calculate the minimum and maximum values in a NumPy array.

Use the `min()` and `max()` functions to calculate the minimum and maximum values in a NumPy array, respectively.

RESULT:

Thus, the Python program successfully implemented aggregation operations using the NumPy library. The program correctly computed the sum, mean, maximum, minimum, standard deviation, variance, and product of the given array elements. The program was executed successfully, and the output was verified.

INFERENCE:

The experiment demonstrates that NumPy aggregation functions provide an efficient and reliable way to perform statistical and numerical computations on arrays. The successful execution of the program validates the use of NumPy for analyzing datasets by accurately computing summary statistics such as sum, mean, maximum, minimum, standard deviation, variance, and product, making it a powerful tool for scientific, engineering, and data analysis applications.

Assessment	Max. Marks	Marks Obtained
Conduction of Practical	15	
Report	5	
Viva	5	
Total Marks	25	
Signature of Lab In-Charge		

